EXAMINATION NO.:_____

THE MALAWI NATIONAL EXAMINATIONS BOARD

2021 MALAWI SCHOOL CERTIFICATE OF EDUCATION EXAMINATION

BIOLOGY

Subject Number: M022/

Tuesday, 2 November

Time Allowed: 2 hour
8:00 – 10:00 an

PAPER I

(100 marks)

Instructions

1. This paper contains **12** printed pages. Please check.
2. Before you begin, **fill** in your **Examination Number** at the top of the question paper and on all other sheets.
3. This paper has **two** sections, **A** and **B**. Section **A** has **ten** questions and Section **B** has **three** questions.
4. Answer all the **13** questions in the spaces provided. The maximum number of marks for each answer is indicated against each question.
5. In the table provided on this page, **tick** against the question number you have answered.
6. At the end of the examination, hand in your paper to the invigilator when time is called to stop writing.

Question Number	Tick if answered	Do not write in these columns	
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			

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Section A (70 marks)

Answer **all** the **ten** questions in this section.

1. **Figure 1** is a diagram showing a plant cell structure as seen under an electron microscope. Use it to answer questions that follow.

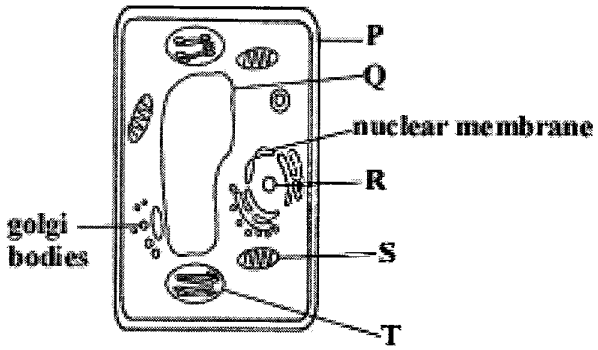

Figure 1

a. Identify the parts labelled **R** and **T**.

R:_____

(1 mark)

T:_____

(1 mark)

b. State the function of part labelled **Q** .

_____

_____

(1 mark)

c. What type of membrane is the part labelled **P**?

_____

(1 mark)

d. Explain the adaptation of the part labeled **S** to its function.

_____

_____

_____

(2 marks)

Continued/…

2. a. A group of Form 3 learners sampled fish in a pond. They obtained 28 fish in the first catch and they marked them. In their second catch they had 15 fish from which 7 were marked. Calculate the estimated population of fish in the fish pond.

(3 marks)

b. State the method used to estimate the population of fish.

_____

(1mark)

c. State any **two** limitations of the method used.

(i) _____

(1 mark)

(ii) _____

(1 mark)

3. a. State any **two** causes of air pollution.

(i) _____

(1 mark)

(ii) _____

(1 mark)

b. Explain how acid rain is caused.

_____

_____

_____

(3 marks)

Continued/…

4. The table below shows frequently eaten foods at household level. Use it to answer questions that follow.

Type of food	Fish	Chicken	Vegetables	Soya Pieces
Frequency	15	10	30	15

a. Draw a bar chart to present the information.

(5 marks)

Continued/…

4. (Continued)

b. Find the modal food.

_____

(1 mark)

5. a. State any **two** functions of the exoskeleton in invertebrates.

_____

(1 mark)

_____

(1 mark)

b. State the **two** differences between Annelids and Nematodes.

Annelids	**Nematodes**
(i)	
(ii)	

(2 marks)

6. a. **Figure 2** is a diagram showing a micro-organism. Use it to answer the questions that follow.

Figure 2

(i) Identify the type of micro-organism.

_____

(1 mark)

(ii) On which subtrate can the micro-organism grow?

_____

(1 mark)

(iii) State any **two** uses of the micro-organism.

_____

(1 mark)

_____

(1 mark)

Continued/…

6. (Continued)

b. (i) Describe the causes of each of the following eye defects.

(1) cataract _____

_____

(2 marks)

(2) long sightedness _____

_____

(2 marks)

(ii) Mention any **two** causes of deafness.

_____

_____

(2 marks)

7. **Figure 3** is a diagram showing a stage in mitosis. Use it to answer questions that follow.

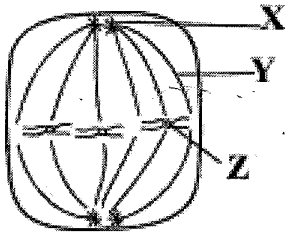

Figure 3

a. Identify the stage of mitosis as shown in **figure 3.**

_____

(1 mark)

b. Name the parts labelled **X** and **Z.**

X: _____

(1 mark)

Z: _____

(1 mark)

c. State any **one** function of part labelled **Y.**

_____

_____

(1 mark)

Continued/…

8. a. State any **two** conditions that facilitate the occurrence of athlete's foot in humans.

_____

_____

(2 marks)

b. Give any **two** ways of preventing the occurrence of athlete's food disease in humans.

_____

_____

(2 marks)

c. Explain any **two** ways how alcohol affects central nervous system in humans.

_____

_____

_____

_____

(4 marks)

9. a. **Figure 4** is a diagram showing a white blood cell. Use it to answer questions that follow.

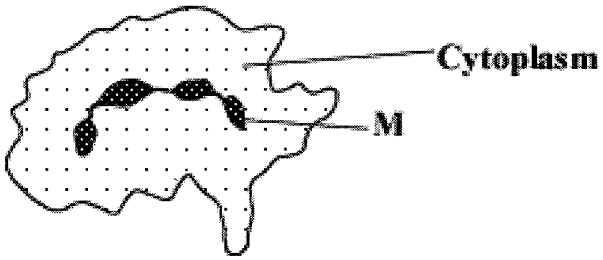

Figure 4

(i) Identify the blood cell.

_____

(1 mark)

(ii) Mention the part labelled **M.**

_____

(1 mark)

Continued/…

9. (Continued)

(iii) Explain briefly how the cell works to defend the body from germs.

_____

_____

_____

_____

_____

(5 marks)

b. (i) Explain any **one** importance of saliva in digestion.

_____

_____

(2 marks)

(ii) Describe any **one** property of enzymes.

_____

_____

(2 marks)

10. a. (i) State any **two** hormones that are involved in the menstrual cycle.

_____

(1 mark)

_____

(1 mark)

(ii) Mention the term used to describe the periodic shedding of the uterine lining through the vagina.

_____

(1 mark)

(iii) Explain what happens to the endometrium soon after the proces mentioned in **(ii)** above.

_____

_____

_____

_____

(2 marks)

Continue

10. (Continued)

b. **Figure 5** is a diagram showing cross-section of a kidney. Use it to answer questions that follow.

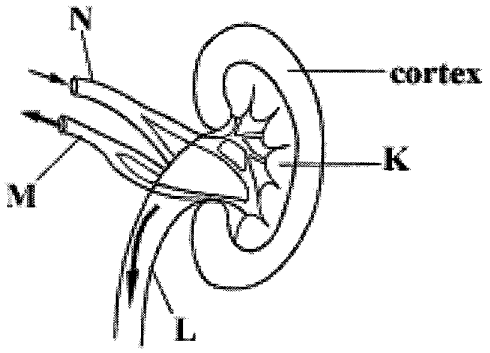

Figure 5

(i) Identify parts labelled **K** and **M**.

K: _____

(1 mark)

M: _____

(1 mark)

(ii) Name the fluid that flows in part labelled **L**.

_____

(1 mark)

(iii) State any **two** differences in the composition of fluid found in **M** and **N**.

_____

_____

_____

_____

(2 marks)

Section B (30 marks)

Answer all the **three** questions in this section. All your answers should be in an essay form.

11. Explain how carbon monoxide poisoning occurs.

(10 marks)

Continued/…

12. Explain any **five** first line defense mechanisms.

(10 marks)

Continued/…

13. Explain the mechanism of evolution by natural selection.

(10 marks)

END OF QUESTION PAPER

NB: This paper contains 12 printed pages.